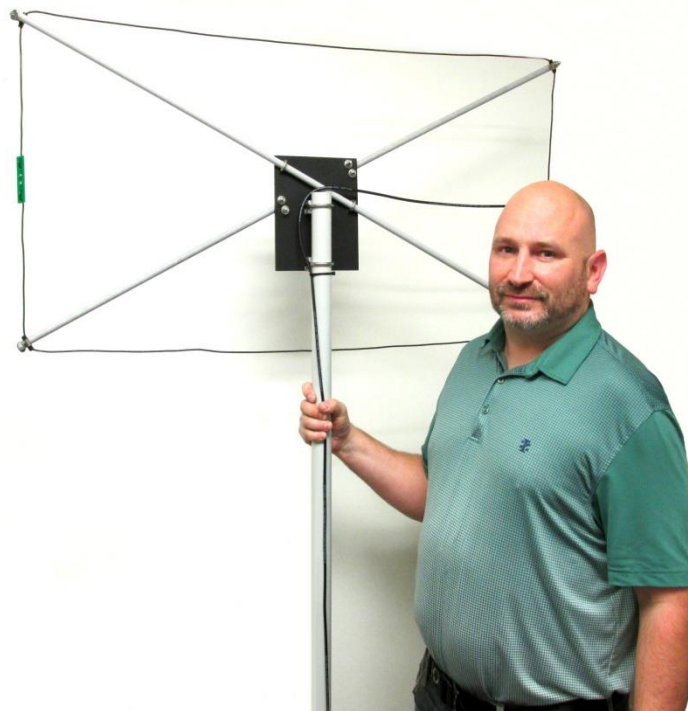




Receive Portable Flag Antenna Kit

DXE-NOISELOOP

DXE-NOISELOOP-INS-Series Rev 0c



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Introduction

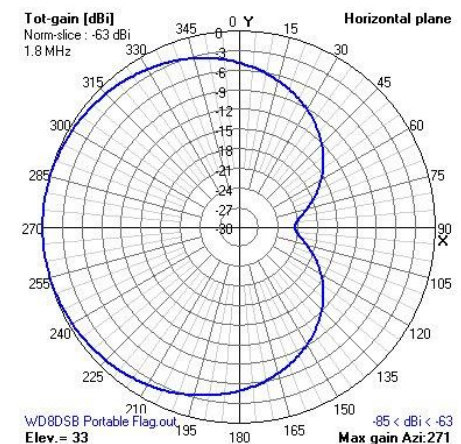
The DX Engineering Receive Portable Flag Antenna kit is based on the design by Don Kirk, WD8DSB. Don's article on this antenna is published in the March, 2021 issue of QST Magazine.

The Portable Flag has become a real game changer in direction finding of RFI on MF and HF frequencies, and you will also find it an invaluable tool when hunting RFI due to the following key features.

- Unidirectional (unlike a tuned loop)
- Broadband (unlike a tuned loop)
- Portable (easy to walk with while direction finding)

Don spent an enormous amount of time tracking down RFI (radio frequency interference), and typically used tuned loops when tracking down MF (medium frequency) and HF (high frequency) signals. Tuned loops work great, but they're bi-directional and therefore the true direction of the signal is unknown due to the 180 degree uncertainty of its bi-directional pattern. Don had been considering a tuned loop along with a sense antenna to make his radio noise direction finding system unidirectional but he felt there must be an easier system to implement. Hence the Receive Portable Flag Antenna, a terminated loop was born.

Traditionally Flags are large and designed for stationary use, whereas the idea for this was to design a Portable Flag. The Portable Flag was designed to be the largest possible (to maximize gain) that would fit inside a car and it's still small enough to maintain a cardioid pattern up through 30 MHz. The portable Flag has an approximate dimension of 2 feet tall by 4 feet long which you (or a helper) can easily walk with while tracking down a signal. Unlike a tuned loop, the Portable Flag is a broadband antenna with unidirectional properties, and it's designed for direction finding from 1.8 to 30 MHz as long as an appropriate preamp is used. It's even useful for direction finding down in the AM broadcast band. The rear of the antenna produces a sharp null when pointed at the signal of interest and the sharp null is often used for direction finding versus using the broad peak which occurs when the front of the antenna is pointed at the signal of interest.



Important note about nearby objects altering the pattern of the antenna:

Skewing of the pattern can happen if you are located close to other metallic objects that can reflect or re-radiate the signal or cause part of the antenna to couple to the metallic object. Always make sure you use the portable flag away from other objects to avoid pattern skewing and this is typical advice when using any kind of direction finding antenna. Use the portable flag when out in the clear for best results. It's very easy for the null to get skewed and when it does not agree with the peak direction, the peak becomes much more reliable for direction finding.

NOTE: This is a **Receive Only** antenna - do **NOT** attempt to transmit through this antenna. Damage to the loop antenna and/or your transmitter may occur.

Optional Items - not included

A custom designed preamplifier / attenuator is available - **DXE-NL-PRE-ATT-1**.

Anti-Seize for stainless steel hardware: Jet-Lube SS-30 **JTL-12555** or Permatex **PTX-81343**

Safety Considerations

WARNING!

USING ANY ANTENNA NEAR POWER LINES IS DANGEROUS



Warning: Do not use the Receive Portable Flag Antenna near overhead power lines or other electric light or power circuits, or where it can come into contact with such circuits. When using the antenna, take extreme care not to come into contact with such circuits, because they may cause serious injury or death.

Parts List

QTY	Description
13 ft	Antenna Wire, 14 ga, Black Relaxed PVC Insulation
1	Binocular Core, Mix 73, 0.3H x 0.4W x 0.6L
1	Coax Cable, RG-58AU, BNC Male Connector Installed
6	DX Engineering Hardware - includes the following: 2 - Hex Nut - Stainless Steel, 1/4-20 2 - Flat Washer - Stainless Steel, 1/4" 2 - Split Washer - Stainless Steel, 1/4"
2	DX Engineering Saddle, 1-1/2"
4	Element Clamp with stud, Stainless Steel, 1/2"
2	Fiberglass Tube, 1/2" OD, .120" wall, 4 feet long
1	Fiberglass Tube, 1-1/2" OD, .120" wall, 4 feet long
4	Hex Nut, Nyloc, Stainless Steel, 10-24
2 ft	Magnet Wire, Green, 28 ga, Nysol
1 ft	Magnet Wire, Red, 28 ga, Nysol

QTY	Description
1	PCB, Feed Point
1	PCB, Resistor
4	P-Clamp, Stainless Steel, 0.3125"
1	Resistor, 680 ohm, 1/4 w
1	Shrink Tube with Adhesive, 3/8" OD, 1.5" Length
1	Starboard Center, Black, Drilled, 6.5" x 8" x 1/4" thick
1	Tape, Double Sided, approx. 0.3" Wide x 0.5" Long
1	Tube, Teflon, 13 AWG, .076" ID x 1.2" long
4	Tube, Wire Guide, 5/16" OD x 2" Long
4	U-Bolt, Stainless Steel, 1/2", 1/4-20 x 0.5 Thread Length
2	U-Bolt, Stainless Steel, 1-1/2", 1/4-20 x 0.5 Thread Length
4	Washer, Flat, Thin, Stainless Steel, #10 x 1/2"
4	Zip Tie, 11" Long
1	Instruction Manual

This kit will typically take between 1 and 2 hours to build.



Assembling the Receive Portable Flag Antenna

1. Assemble the transformer core using the following components:
Binocular Core, Teflon Tube, Green Magnet Wire and Red Magnet Wire.

The core is made into a transformer that uses the Green wire as the Secondary and the Red wire as the Primary,



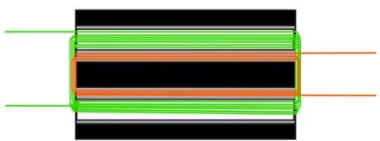
2. Using the core as a guide, cut the Teflon tube into 2 pieces the same length as the core.



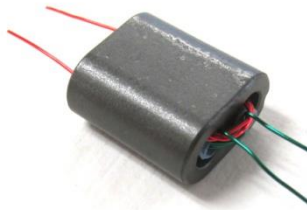
3. Using the green magnet wire, insert one end into one of the Teflon tubes in the core allowing approximately 1/2" hanging out. Make **12 turns** through the Teflon tubes in the core. Keep the wire snug. One turn is once through each side of the core.

Using the red magnet wire, insert one end into the core (**not through the Teflon tubes**) allowing approximately 1/2" hanging out. Make **3 turns** through the core.

Trim the wires so there is approximately 1/2" sticking out both sides of the core.



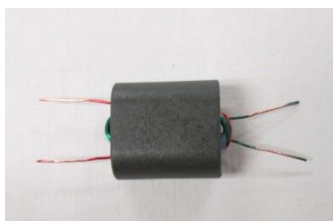
Cut away view showing Green wires going through Teflon tubes



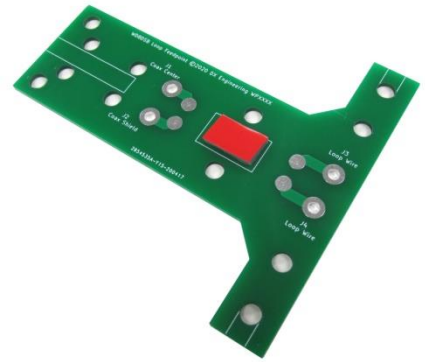
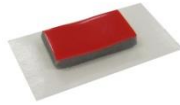
Red is Primary

Green is Secondary

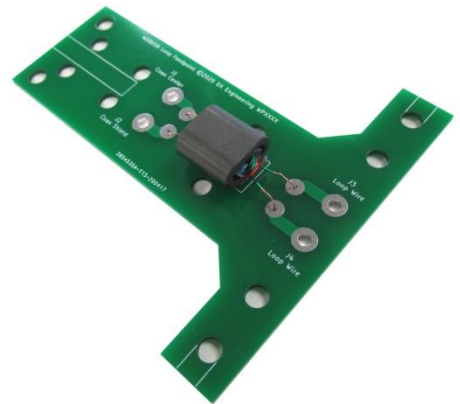
4. Use a sharp knife to carefully scrape the enamel coating off of the ends of the wires as shown. Take care not to nick or cut the wires.



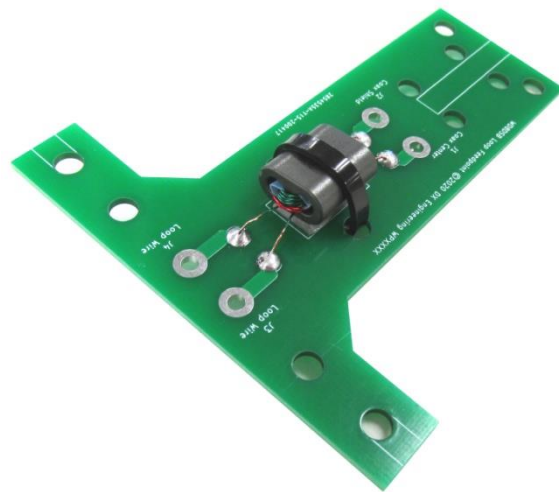
5. Attach the double sided tape to the feed point PCB, as shown. Then remove the top paper exposing the adhesive.



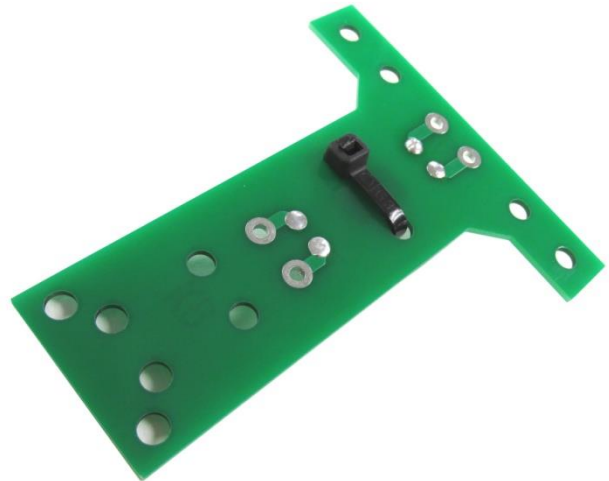
6. Press the transformer on the double sided tape on to the feed point PCB as shown. **Note colors of the wires in photo.** The green wires (secondary) connect to the “Loop Wire” holes, the red wires (primary) connect to the holes near the Coax connections.



7. Solder the wire leads in place. Trim wires.



8. Use one zip tie to hold the transformer in place on the feed point PCB as shown. Do not over tighten the zip tie.



9. Using the 13 feet of the black antenna wire, cut it in half to make 2 wires the same length. Strip approximately 1/4" to 3/8" off one end of each wire.

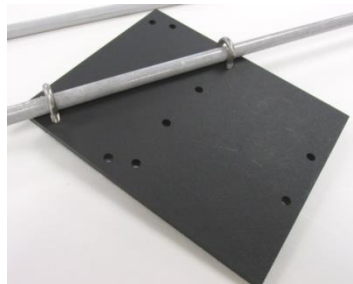
10. Route one wire through the resistor PCB as shown and trim as needed. Route the **second** wire through the resistor PCB as shown, solder the two ends in place and trim as needed.



11. Install the 680 ohm 1/4 watt resistor in the resistor PCB, solder in place and trim as needed.

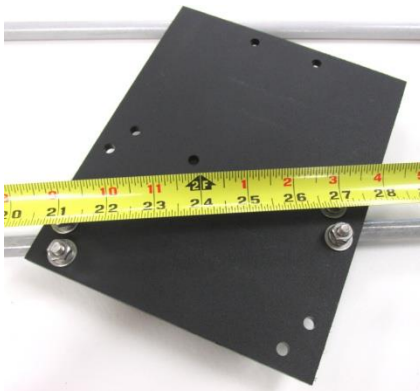


12. Assemble one 1/2" OD cross fiberglass rod to the black starboard center board as shown. Once the U-bolts are installed, just snug, do not tighten them at this time.



13. To center up the fiberglass rod, use a tape measure and measure one half the length of the rod from the end of the fiberglass rod to the center of the starboard center board. Once the rod is centered, tighten the U-bolt hardware for that rod. Do not over tighten. The distance from the end of one rod to the U-bolt hardware is 21 inches as shown. From the other end of hat same rod, it should also be 21 inches to the U-bolt hardware on that side as well. Evenly tighten the u-bolts in place.

14. Repeat for the second fiberglass rod on the other side of the starboard center as shown.



15. Using the element clamp with mounting stud, install one to the each of the ends of the 1/2" OD fiberglass rods as shown. **Note the direction of the mounting studs. One set is the opposite direction than the other.** This keeps the antenna wire centered front to back. Tighten the element clamps in place.



16. Install one p-clamp, wire guide, flat washer and Nyloc hex nut onto each of the element stud clamps as shown. Tighten in place.



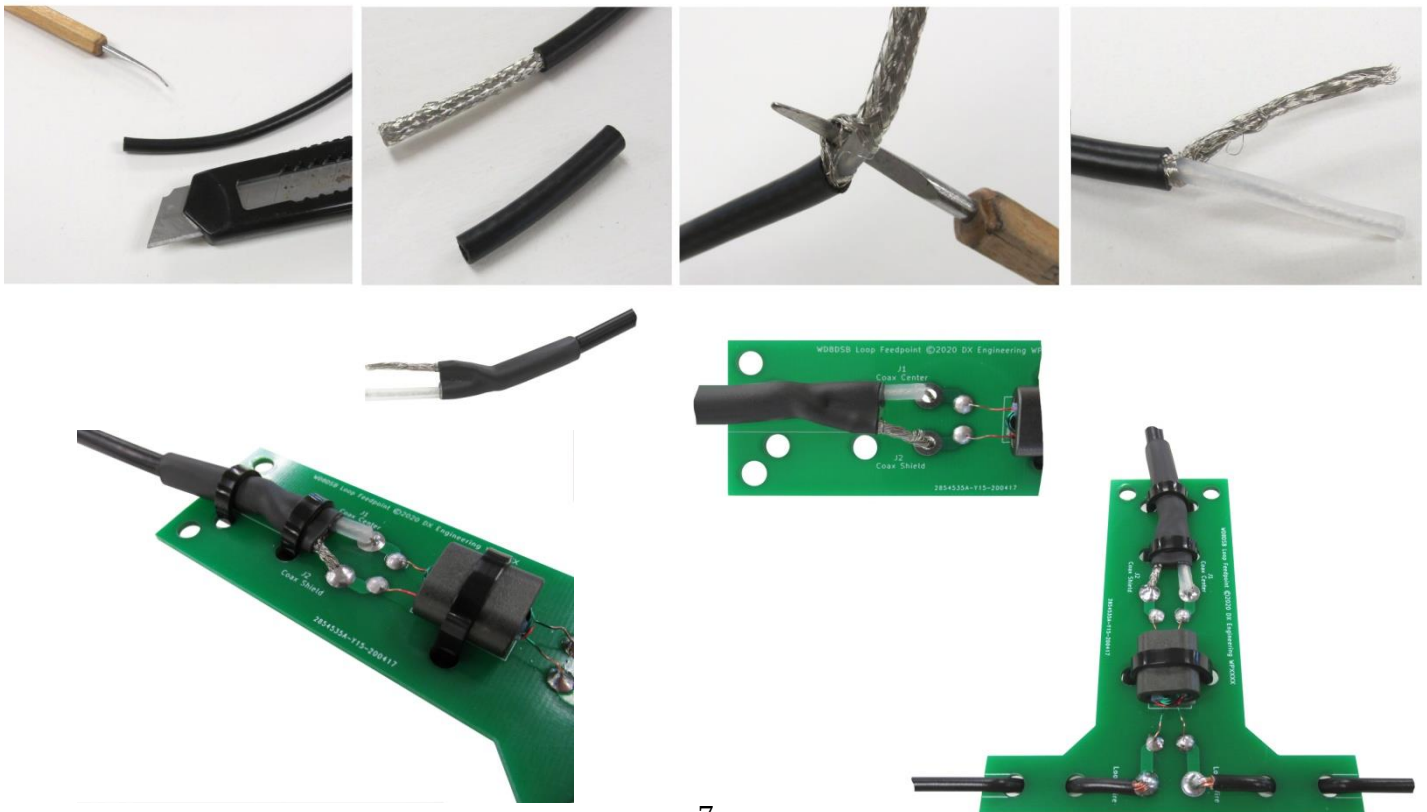
17. Take the resistor PCB that has the two antenna wires soldered in place and run each wire through the wire guides as shown. **Note the position of the center board.** Keep the resistor PCB **centered** between the two rods as shown.



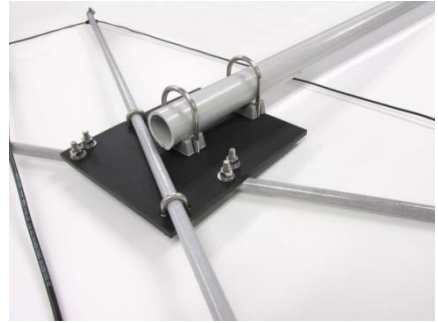
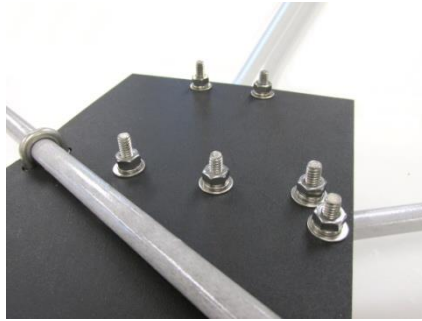
18. Use the feed point PCB on the other side of the loop and route the antenna wires through the guides and then through the feed point PCB **keeping the feed point PCB centered between the rods**. *Ensure the resistor PCB is still also centered*. Snug up the wires and make a small mark on each wire just above where it passes through the PCB eyelets that will be soldered. Remove the wires from the feed point PCB, trim the wire lengths and insulation back 1/4" to 3/8" on the ends. Re-route the wire through the feed point PCB **ensuring all is centered and snug**. Solder the two wires on the feed point PCB.



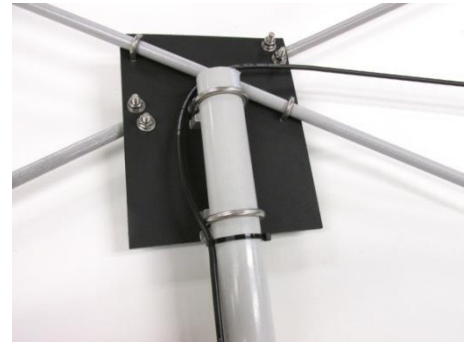
19. Prepare the coaxial cable with BNC connector as shown below. Cut approximately one inch of the black insulation off of the coax end. Take care not to cut or nick the shield. You want to separate the shield and center insulation as shown using a soldering pick or other small tool (small Phillips head screwdriver works good). Trim about 1/2" of the center insulation from the center coax conductor. Take care not to cut or nick the center conductor strands. Install the shrink tube with adhesive near the ends of the prepped coax using a heat gun or other hot air source to shrink the tube as shown. Insert the shield and the center conductor of the coax assembly through the feed point PCB and solder in place. Use two zip ties to snugly hold the coax cable against the feed point PCB as shown.



20. Install the 1-1/2" fiberglass rod (handle for the antenna) to the center board using the saddle clamps as shown below. Evenly tighten the U-Bolts in place. Do not over tighten.



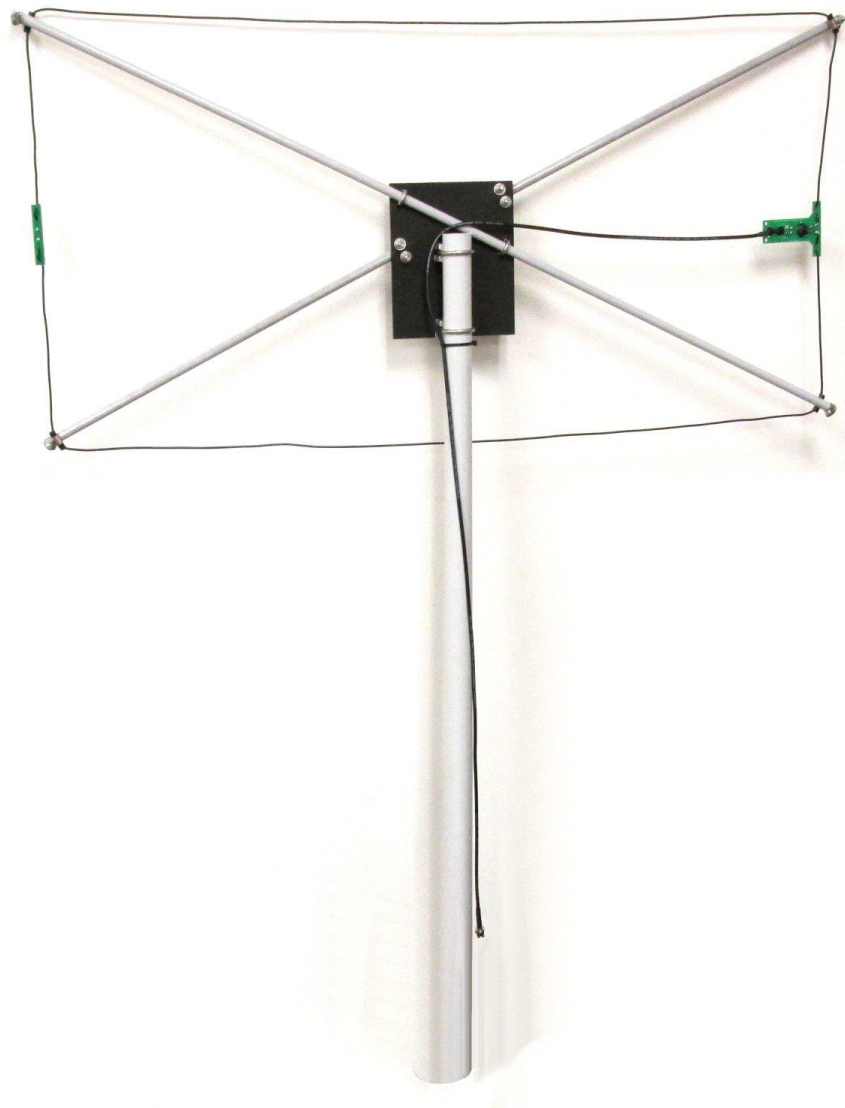
21. Route the coaxial cable as shown and use a zip tie to hold it in place. Do not over tighten the zip tie, you just want it snug.



22. Use an antenna analyzer to verify the antenna is properly assembled and tunes and there is no shorted coaxial cable. The reading is typically 1:1 to 2:1 when swept as shown in the photo.



Overall View of completed Receive Loop antenna



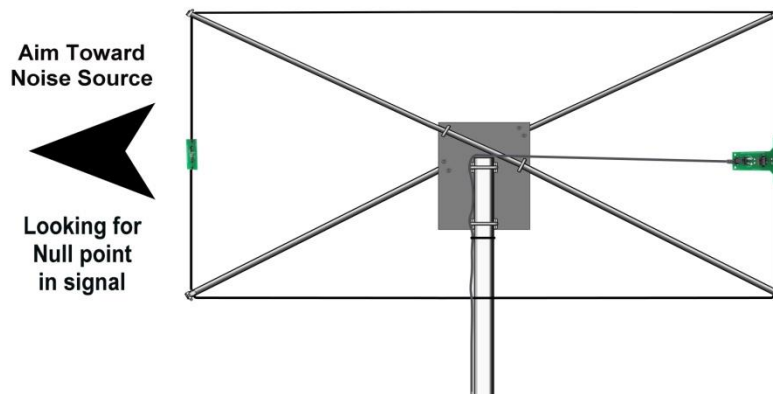
Optional Pre-Amplifier and Attenuator

A perfect companion to the DXE-NOISELOOP is the optional Pre-Amplifier/Attenuator **DXE-NL-PRE-ATT-1**. It uses an internal 9 volt battery, has BNC connectors for Input and Output, Easy to use slide switches and a power on LED when the Pre-Amp is turned on. Selectable attenuations levels of -10 db, -20 db (and -30 db when both are on), plus an additional AM filter can be switched in when needed. Note that this is a Receive Only system. Transmitting RF will damage the unit and it is not covered by warranty.



Using the DXE-NOISELOOP

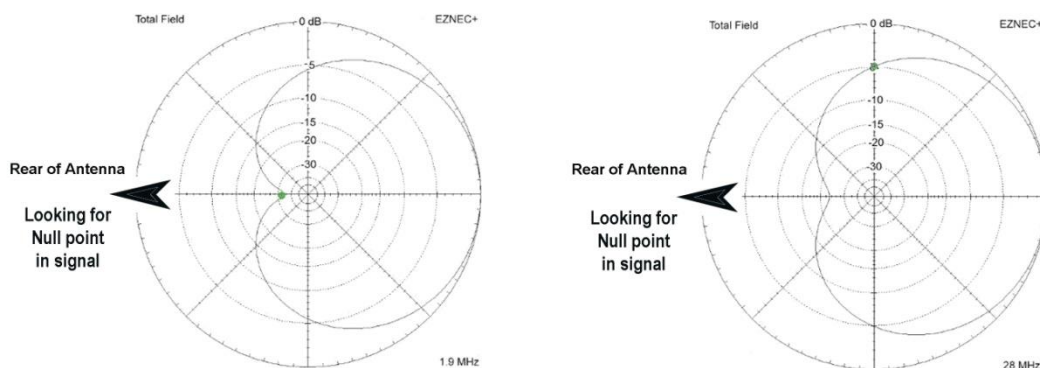
The DXE-NOISELOOP is directional. The **rear** of the antenna is toward the smaller resistor PCB. This would be the maximum **null** point (unlike a regular antenna where you are normally looking for a peak signal). The front of the antenna is by the larger PCB that holds the transformer. When tracking down an offending signal, using the null will allow you an easier method to determine the direction.



When testing was being done at DX Engineering, an ICOM IC-705 was used because that radio is portable, easy to use and has the ability to display the selected spectrum display as an added feature when tracking down an offending HF signal.

Holding the DXE-NOISELOOP vertically with the radio attached and tuned to the HF band desired, rotate the antenna until there is a **null** (not the peak) in the selected, desired noise frequency.

Moving 90 degrees away from your current position, then turning the antenna again to null the desired noise frequency, will allow you to triangulate the spot that the interfering signal is coming from. The rear of the antenna produces a sharp null when pointed at the signal of interest and the sharp null is often used for direction finding versus using the broad peak which occurs when the front of the antenna is pointed at the signal of interest.



Plots showing 1.9 MHz and 28 MHz - note the null points

If you have a problem with AM broadcast adding additional interference, the optional **DXE-NL-PRE-ATT-1** Noise Loop has extra filtering built in. When the AM Filter is turned on, you will see a large decrease from offending AM Broadcast signals.

Manual Updates and Information

Every effort is made to supply the latest manual revision with each product. Occasionally a manual will be updated between the time your DX Engineering product is shipped and when you receive it. Please check the DX Engineering web site (www.dxengineering.com) for the latest revision manual.

Technical Support

If you have questions about this product, or if you experience difficulties during the installation, contact DX Engineering at (330) 572-3200. You can also e-mail us at: DXEngineering@DXEngineering.com
For best service, please take a few minutes to review this manual before you call.

Warranty

All products manufactured by DX Engineering are warranted to be free from defects in material and workmanship for a period of one (1) year from date of shipment. DX Engineering's sole obligation under these warranties shall be to issue credit, repair or replace any item or part thereof which is proved to be other than as warranted; no allowance shall be made for any labor charges of Buyer for replacement of parts, adjustment or repairs, or any other work, unless such charges are authorized in advance by DX Engineering. If DX Engineering's products are claimed to be defective in material or workmanship, DX Engineering shall, upon prompt notice thereof, issue shipping instructions for return to DX Engineering (transportation-charges prepaid by Buyer). Every such claim for breach of these warranties shall be deemed to be waived by Buyer unless made in writing. The above warranties shall not extend to any products or parts thereof which have been subjected to any misuse or neglect, damaged by accident, rendered defective by reason of improper installation, damaged from severe weather including floods, or abnormal environmental conditions such as prolonged exposure to corrosives or power surges, or by the performance of repairs or alterations outside of our plant, and shall not apply to any goods or parts thereof furnished by Buyer or acquired from others at Buyer's specifications. In addition, DX Engineering's warranties do not extend to other equipment and parts manufactured by others except to the extent of the original manufacturer's warranty to DX Engineering. The obligations under the foregoing warranties are limited to the precise terms thereof. These warranties provide exclusive remedies, expressly in lieu of all other remedies including claims for special or consequential damages. SELLER NEITHER MAKES NOR ASSUMES ANY OTHER WARRANTY WHATSOEVER, WHETHER EXPRESS, STATUTORY, OR IMPLIED, INCLUDING WARRANTIES OF MERCHANTABILITY AND FITNESS, AND NO PERSON IS AUTHORIZED TO ASSUME FOR DX ENGINEERING ANY OBLIGATION OR LIABILITY NOT STRICTLY IN ACCORDANCE WITH THE FOREGOING.

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DXE-NOISELOOP-INS